

Dilawar Singh

✉ dilawar.s.rajput@gmail.com • 🌐 dilawars.me • 🐙 [dilawar](https://github.com/dilawar)

Versatile system engineer (Rust, C++) with significant tech lead experience looking for deep technical roles in a structured environment. Experienced at modelling and developing digital systems, close-to-kernel application, simulators, and algorithms.

Experience

Senior Engineer, Dognosis, Bengaluru

Jan 2025 – current

- Designed and built data APIs (Rust) and portals (PHP) to collect live data from various sensors and agents
- Added SLAM to canine kit to create geo-spatial model, unlocking couple of use-cases in threat detection
- Setup hiring pipelines, interviewed and hired 5 engineers to increase team sizes to 11 in 3 months

Cofounder & CTO, Subconscious Compute, Bengaluru

Dec 2019 – Dec 2024

- Cofounded a endpoint cybersecurity startup and took it through pre-series A. Hired and led a team of 15 system engineers to build our flagship endpoint security product Shepherd
- Wrote kernel minifilter (C++, Rust), designed core algorithms (Rust), and setup devops for cross-platform deployment. Wrote core engine for threat detection in network and data streams generated by eBPF sensors.

Research Fellow & GSoC Mentor

2016 – 2018

- Added multiple numerical solvers (C++, Boost, GSL) to [MOOSE](#) that allowed broader use in multiscale problems
- Mentored GSoC participants of INCF for 3 years on GPU/CUDA effort resulting in faster simulations

Firmware Engineer, Kritikal Solutions, Noida

Jul 2009 – Jun 2010

- Implemented Kalman filter based image stabilizer that removed most vibration artifacts from recorded videos

Projects

- Built a [Behaviour Box](#) using (C++, Python, BlackFly Camera, Raspberry Pi, Custom Hardware) that automated training session making multiple sessions possible in a day!
- Wrote and maintain a few [rust crates](#) on stream processing, WMI interface, and a few cli utilities
- Wrote and maintain a few [python packages](#): a popular utility to extract [data from images of plots](#), Notion/Gitlab bridge for internal use, a UI for oscilloscope and a simple plagiarism detector [CodeSniffer](#) as teaching assistant
- Wrote python bindings and maintain [wheels](#) for two major C++ simulators [Smoldyn](#) and [MOOSE](#)

Skills

Languages: Rust, C++, Python, SQL, $\text{\LaTeX}$. (*familiar*) Haskell, SystemC, Verilog, VHDL.

Libraries : Boost, Win32 API, PyBind11, OpenCV; Numpy/Scipy, Pandas, librosa;

Simulation & Modeling: KiCAD, OpenSCAD; ModelSim, NgSpice; MOOSE, Neuron, Smoldyn; PlantUML, TikZ;

Tech & Ops: Git, Docker, CMake, PyInfra/Ansible, Shell scripting/Lua; Jenkins/Gitlab Runners; Packaging (nsis, deb, rpm), [Open Build Service](#);

Databases: PostGreSQL, MariaDB, Sqlite3, DuckDB, VictorialMetrics

Misc: Literature survey and analysis, Parsing, Network Flows, Graphs, Linear Algebra, Combinatorics, Computer Vision, Probabilistic DSA, BDD & Model Checking, DSP, Routing;

Education

PhD, Computational Neuroscience, NCBS Bangalore (TIFR Mumbai), India

2019

PhD, Digial Systems, IIT Bombay, **withdrawn**

2013

Master of Technology, Electrical Engineering (VLSI), IIT Bombay

2009

BTech, Instrumentation and Control Engineering, Dr. MGR ERI, Chennai

2007